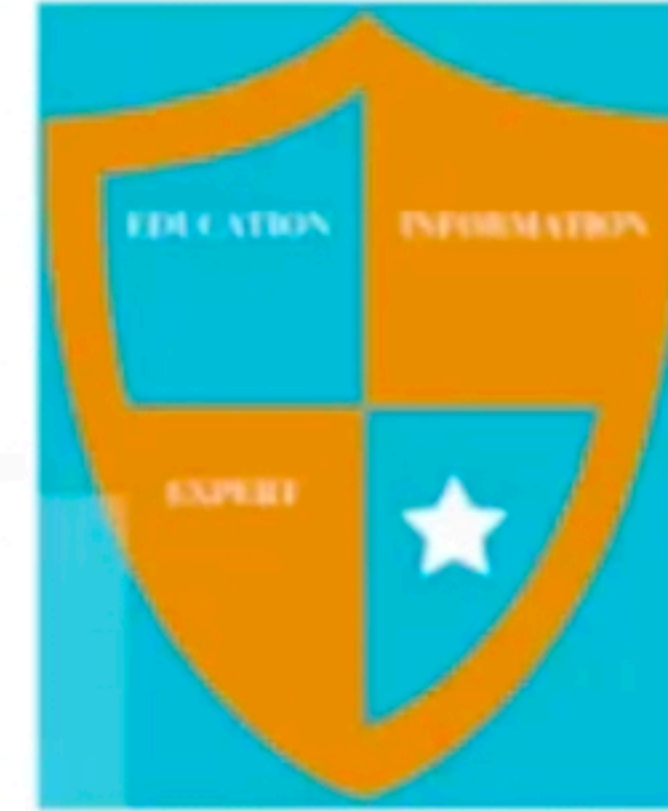


Question 1

Show that $(\alpha T)^* = \alpha^{-1} T^*$



Solution:

$$\langle x, (\alpha T^*)^* y \rangle = \langle (\alpha T)x, y \rangle$$

$$\langle x, (\alpha T^*)^* y \rangle = \langle \alpha(Tx), y \rangle$$

$$\langle x, (\alpha T^*)^* y \rangle = \bar{\alpha} \langle Tx, y \rangle$$

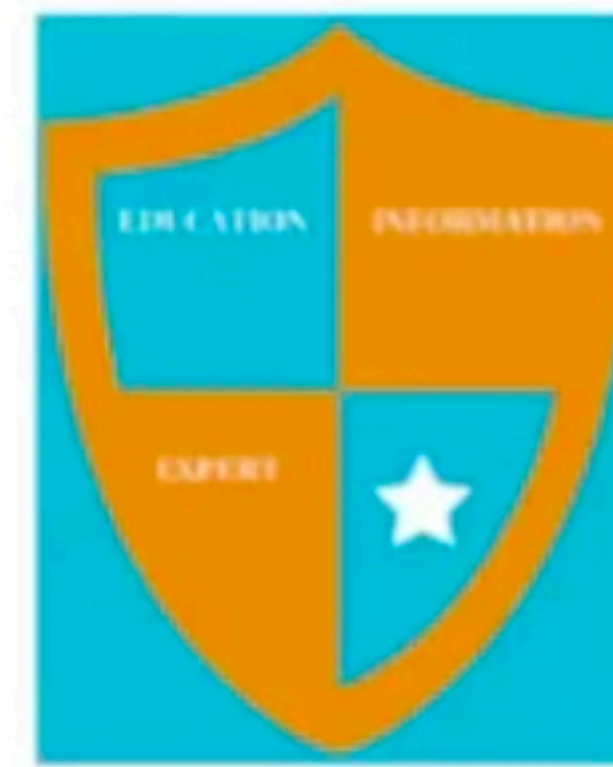
$$\langle x, (\alpha T^*)^* y \rangle = \bar{\alpha} \langle x, T^* y \rangle$$

$$\langle x, (\alpha T^*)^* y \rangle = \langle x, \bar{\alpha} T^* y \rangle$$

$$(\alpha T^*)^* = \bar{\alpha} T^*$$



Question 2



Prove that $(S + T)^* = S^* + T^*$

Prove:

$$\langle x, (S + T)^* y \rangle = \langle (S + T)x, y \rangle$$

$$\langle x, (S + T)^* y \rangle \geq \langle (Sx + Tx), y \rangle$$

$$\langle x, (S + T)^* y \rangle = \langle Sx, y \rangle + \langle Tx, y \rangle$$

$$\langle x, (S + T)^* y \rangle = \langle x, S^* y \rangle + \langle x, T^* y \rangle$$

$$\langle x, (S + T)^* y \rangle = \langle x, (S^* y + T^* y) \rangle$$

$$\langle x, (S + T)^* y \rangle = \langle x, (S^* + T^*)y \rangle$$

Hence we have proved that

$$(S + T)^* = S^* + T^*$$



Question 3



Find matrix multiplication if $(\xi_1, \xi_2, \xi_3) \leftrightarrow (\xi_1, \xi_2, -\xi_1 - \xi_2)$

Solution:

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \begin{bmatrix} \xi_1 \\ \xi_2 \\ \xi_3 \end{bmatrix} = \begin{bmatrix} \xi_1 \\ \xi_2 \\ -\xi_1 - \xi_2 \end{bmatrix}$$

Hence required matrix is

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & -1 & 0 \end{bmatrix}$$



Question 3



Find matrix multiplication if $(\xi_1, \xi_2, \xi_3) \leftrightarrow (\xi_1, \xi_2, -\xi_1 - \xi_2)$

Solution:

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix} \begin{bmatrix} \xi_1 \\ \xi_2 \\ \xi_3 \end{bmatrix} = \begin{bmatrix} \xi_1 \\ \xi_2 \\ -\xi_1 - \xi_2 \end{bmatrix}$$

Hence required matrix is

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -1 & -1 & 0 \end{bmatrix}$$



Question 4

$$\|x_1 + x_2 + \dots + x_n\|^2 = \|x_1\|^2 + \|x_2\|^2 + \dots + \|x_n\|^2$$

Prove:

$$\left\| \sum_{i=1}^n x_i \right\|^2 = \left\langle \sum_{i=1}^n x_i, \sum_{i=1}^n x_i \right\rangle$$

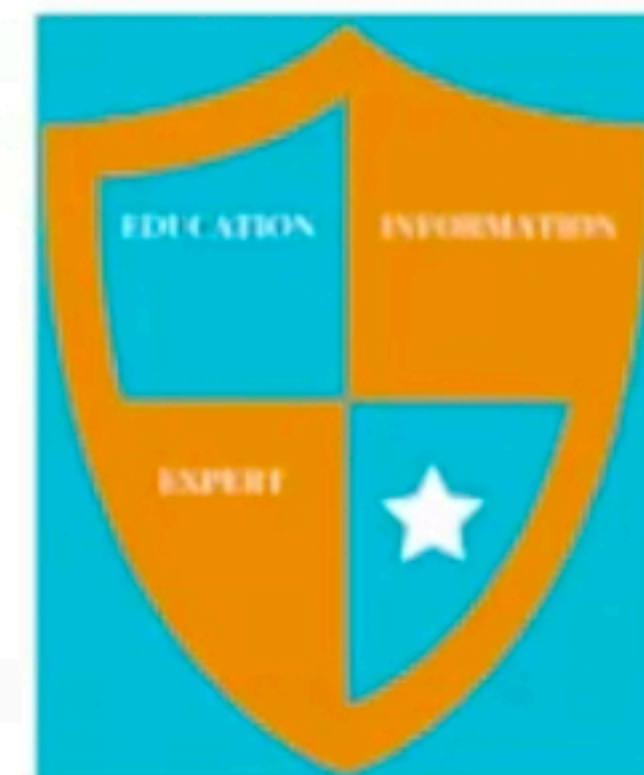
$$= \langle x_1 + \dots + x_n, x_1 + \dots + x_n \rangle$$

$$= \langle x_1, x_1 + \dots + x_n \rangle + \dots + \langle x_n, x_1 + \dots + x_n \rangle$$

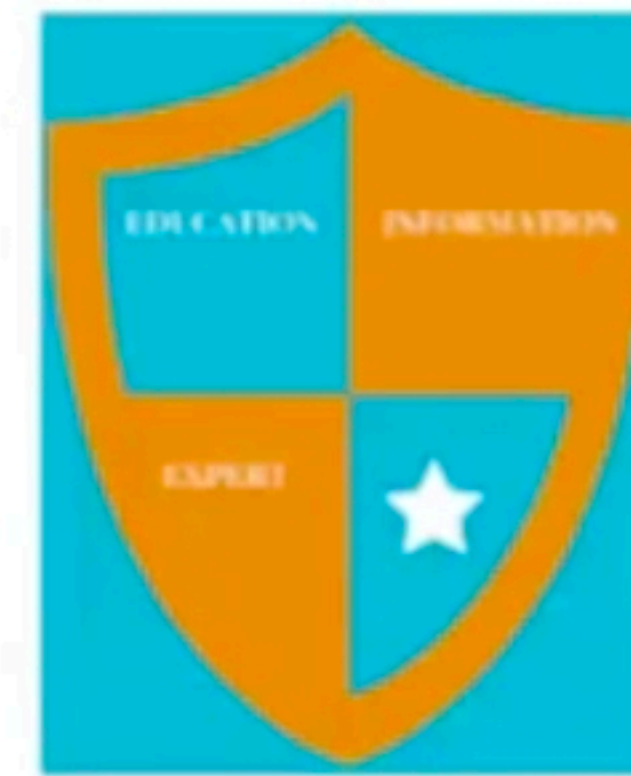
$$= \sum_{i=1}^n \sum_{j=1}^n \langle x_i, x_j \rangle$$

$$\langle x_i, x_j \rangle = \|x_i\|^2, \text{ if } i = j \quad \langle x_i, x_j \rangle = 0 \text{ and for } i \neq j \text{ then } \langle x_i, x_j \rangle = 0$$

$$\left\| \sum_{i=1}^n x_i \right\|^2 = \sum_{i=1}^n \|x_i\|^2$$



Question 5



Prove that $(ST)^* = S^*T^*$

Prove:

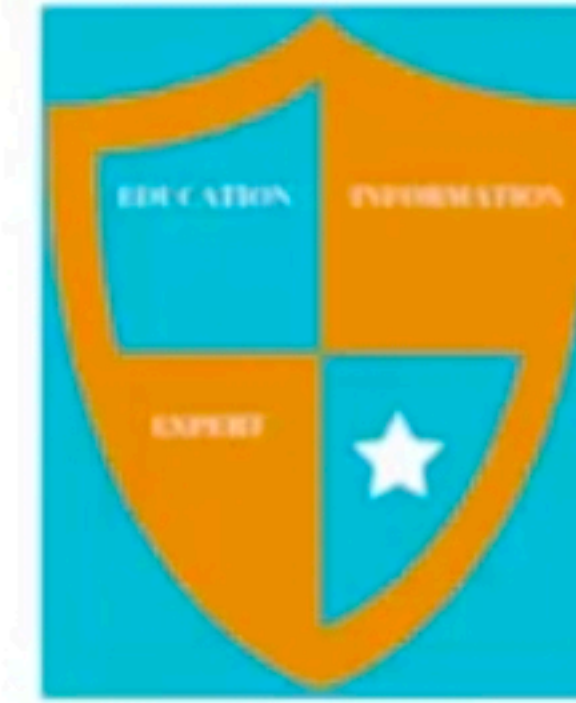
$$\begin{aligned} \langle x, (ST)^*y \rangle &= \langle (ST)x, y \rangle \\ \langle x, (ST)^*y \rangle &\geq \langle (SxTx), y \rangle \\ \langle x, (ST)^*y \rangle &= \langle Sx, y \rangle \langle Tx, y \rangle \\ \langle x, (ST)^*y \rangle &= \langle x, S^*y \rangle \langle x, T^*y \rangle \\ \langle x, (ST)^*y \rangle &= \langle x, (S^*+T^*)y \rangle \\ \langle x, (ST)^*y \rangle &= \langle x, (S^*T^*)y \rangle \end{aligned}$$

Hence we have proved that

$$(ST)^* = S^*T^*$$



Question 6



Show that $\overline{N(T)} < N(T)$

Prove:

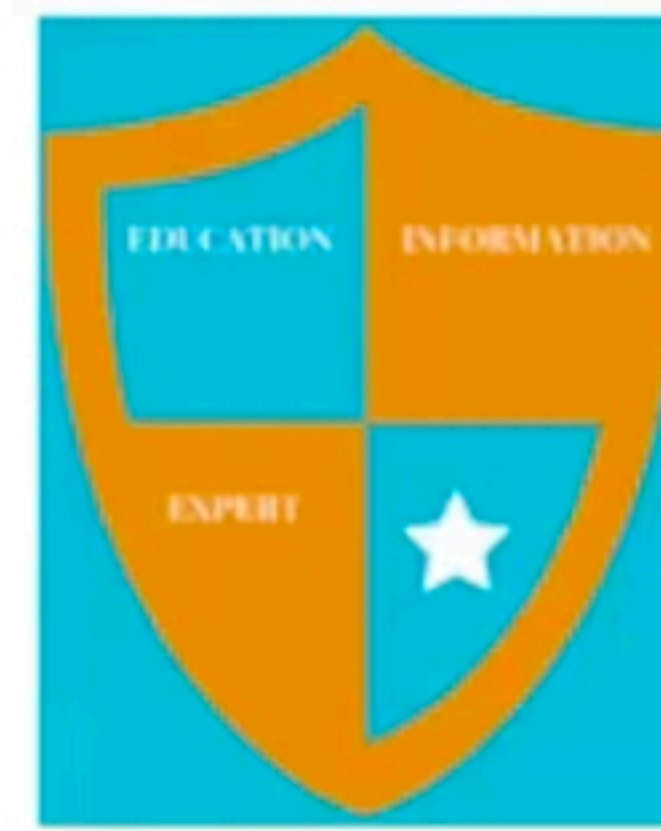
$$\begin{aligned} & x \in N(T) \\ \exists (x_n) \text{ such that } & x_n \in N(T) + x_n \rightarrow x \end{aligned}$$

Therefore

$$\begin{aligned} & Tx_n \rightarrow Tx \\ & x_n \in N(T) \\ & Tx_n = 0 \\ & T_n = 0 \\ & x \in N(T) \\ & \overline{N(T)} < N(T) \\ & \overline{N(T)} = N(T) \\ & N(T) \text{ is closed.} \end{aligned}$$



Question 7



Prove that $\langle x, ay + bz \rangle = \bar{a} \langle x, y \rangle + \bar{b} \langle x, z \rangle$

Prove:

$$\langle x, ay + bz \rangle = \langle \overline{ay + bz}, \bar{x} \rangle$$

$$\langle x, ay + bz \rangle = \langle \bar{a}\bar{y}, \bar{x} \rangle + \langle \bar{b}\bar{z}, \bar{x} \rangle$$

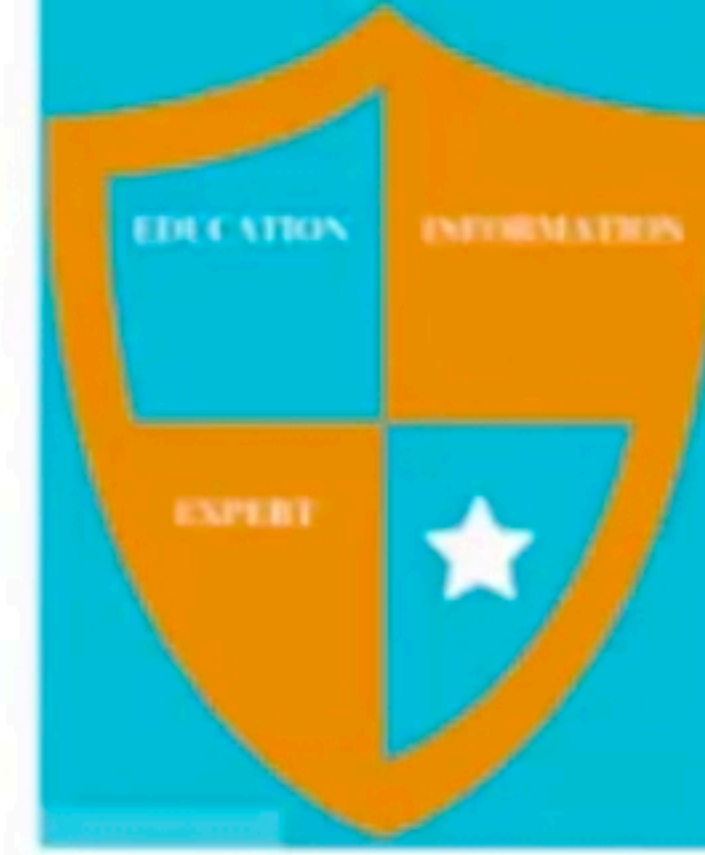
$$\langle x, ay + bz \rangle = \bar{a} \langle \bar{y}, \bar{x} \rangle + \bar{b} \langle \bar{z}, \bar{x} \rangle$$

$$\langle x, ay + bz \rangle = \bar{a} \langle x, y \rangle + \bar{b} \langle x, z \rangle$$

Hence proved.



Question 8



Let H_1 and H_2 be Hilbert spaces, $S: H_1 \rightarrow H_2$ and $T: H_1 \rightarrow H_2$ be bounded linear operator and α any scalar. Then show that $(T^*)^* = T$.

Prove:

$$\langle x, (T^*)^* y \rangle = \langle x T^*, y \rangle$$

$$\langle x, (T^*)^* y \rangle = \langle x, T y \rangle$$

It implies that

$$(T^*)^* = T \text{ so its proved.}$$



Question 9

Prove that the product of two bounded self-adjoint linear operators S and T on a Hilbert space H is self-adjoint if and only if the operators commute, $ST = TS$.

Prove:

$$\begin{aligned} \langle x, (ST)^*y \rangle &= \langle (ST)x, y \rangle \\ \langle x, (ST)^*y \rangle &= \langle (Sx)Tx, y \rangle \\ \langle x, (ST)^*y \rangle &= \langle Sx, y \rangle \langle Tx, y \rangle \\ \langle x, (ST)^*y \rangle &= \langle x, S^*y \rangle \langle x, T^*y \rangle \\ \langle x, (ST)^*y \rangle &= \langle x, (S^*+T^*)y \rangle \\ \langle x, (ST)^*y \rangle &= \langle x, (S^*T^*)y \rangle \end{aligned}$$



Hence we have proved that

$$(ST)^* = S^*T^*$$

$$(ST)^* = T^*S^* = TS.$$

Hence

$$ST = (ST)^* \iff ST = TS.$$



Question 10

Let H_1 and H_2 be Hilbert spaces, $S: H_1 \rightarrow H_2$ and $T: H_1 \rightarrow H_2$ be bounded linear operator and α any scalar. Then show that $\|T^*T\| = \|TT^*\| = \|T\|^2$.

Prove:

$$\|Tx\|^2 = \langle Tx, Tx \rangle = \langle T^*Tx, x \rangle \leq \|T^*Tx\| \|x\| \leq \|T^*T\| \|x\|^2.$$

Taking the supremum over all x of norm 1, we obtain $\|T\|^2 \leq \|T^*T\|$.
we have

$$\|T\|^2 \leq \|T^*T\| \leq \|T^*\| \|T\| = \|T\|^2.$$

Hence $\|T^*T\| = \|T\|^2$. Replacing T by T^* and we also have

$$\|T^{**}T^*\| = \|T^*\|^2 = \|T\|^2.$$

Here $T^{**} = T$



Question 11

Let the operators $U: H \rightarrow H$ and $V: H \rightarrow H$ be unitary; here, H is a Hilbert space. Then show that U is isometric; thus $\|Ux\| = \|x\|$ for all $x \in H$.

Prove:

$$\|Ux\|^2 = \langle Ux, Ux \rangle$$

$$\|Ux\|^2 = \langle x, U^*Ux \rangle$$

$$\|Ux\|^2 = \langle x, Ix \rangle$$

$$\|Ux\|^2 = \langle x, x \rangle$$

$$\|Ux\|^2 = \|x\|^2$$

It proved.



Question 12



Show that $\langle T^*y, x \rangle = \langle y, Tx \rangle$

Prove:

$$\langle T^*y, x \rangle = \langle \overline{y}, \overline{T^*x} \rangle$$

$$\langle T^*y, x \rangle = \langle \overline{Tx}, \overline{y} \rangle$$

$$\langle T^*y, x \rangle = \langle y, Tx \rangle$$

Hence its proved.



Question 13



Let X and Y be inner product spaces and $Q: X \rightarrow Y$ a bounded linear operator. Then Show that $Q=0$ if and only if $\langle Qx, y \rangle = 0$ for all $x \in X$ and $y \in Y$.

Prove:

$$Q = 0 \implies Qx = 0 \text{ for all } x \in X$$

$$\langle Qx, y \rangle = \langle 0, y \rangle$$

$$\langle Qx, y \rangle = \langle 0, w, y \rangle$$

$$\langle Qx, y \rangle = 0 \langle w, y \rangle$$

$$\langle Qx, y \rangle = 0 \text{ for all } x \in X$$

It implies that $Qx=0$ and by definition $Q=0$. Its proved.



Question 14

Let the operators $U: H \rightarrow H$ and $V: H \rightarrow H$ be unitary; here, H is a Hilbert space. Then show that $\|U\|=1$, provided $H \neq \{0\}$.

OR

Let the operators $U: H \rightarrow H$ and $V: H \rightarrow H$ be unitary; here, H is a Hilbert space. Then show that $U^{-1}(=U^*)$ is unitary.

Prove:

Since U is bijective it shows that U^{-1} is bijective

$$(U^{-1})^* = (U^*)^* = U = (U^{-1})^{-1}.$$

Hence it implies that U^{-1} is unitary.



Question 15



Let the operators $U: H \rightarrow H$ and $V: H \rightarrow H$ be unitary; here, H is a Hilbert space. Then show that U is normal.

Prove:

$$UU^* = U^*U$$

Since we know that

$$U^* = U^{-1}$$

So

$$UU^{-1} = U^{-1}U$$

$(UV)^* = (UV)^{-1} = I$ Hence its proved that U is normal.



Question 16



In an inner product space V and x, y in V if $x \perp y$ then

$$\|x + y\|^2 = \|x\|^2 + \|y\|^2$$

Prove:

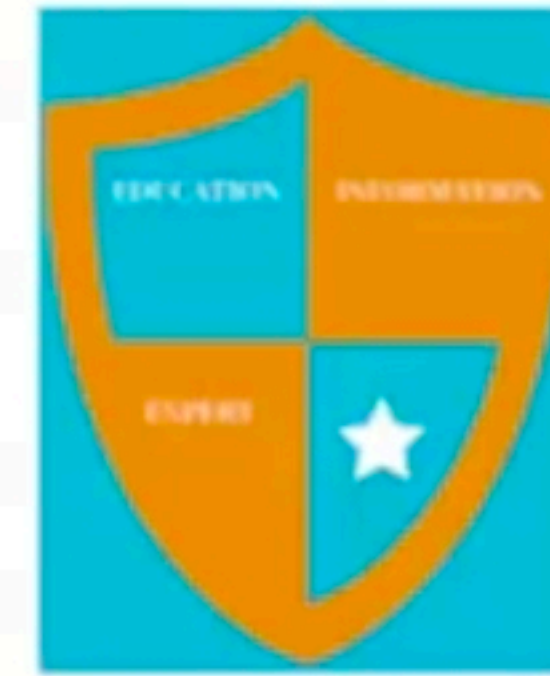
$$\begin{aligned}\|x + y\|^2 &= \langle x + y, x + y \rangle \\ &= \langle x, x \rangle + \langle x, y \rangle + \langle y, x \rangle + \langle y, y \rangle\end{aligned}$$

As x and y are perpendicular so $\langle x, y \rangle = 0, \langle y, x \rangle = 0$

$$\|x + y\|^2 = \langle x, x \rangle + \langle y, y \rangle = \|x\|^2 + \|y\|^2$$



Question 17



$$\|x+y\|^2 + \|x-y\|^2 = 2(\|x\|^2 + \|y\|^2) \quad \text{for all } x, y \in V$$

Prove:

$$\begin{aligned}\|x+y\|^2 &= \langle x+y, x+y \rangle \\ &= \langle x, x \rangle + \langle x, y \rangle + \overline{\langle x, y \rangle} + \langle y, y \rangle \\ &= \|x\|^2 + 2\operatorname{Re} \langle x, y \rangle + \|y\|^2 \quad \dots(i)\end{aligned}$$

Replace $y=-y$

$$\begin{aligned}\|x-y\|^2 &= \langle x-y, x-y \rangle \\ &= \langle x, x \rangle - \langle x, y \rangle - \overline{\langle x, y \rangle} + \langle y, y \rangle \\ &= \|x\|^2 - 2\operatorname{Re} \langle x, y \rangle + \|y\|^2 \quad \dots(ii)\end{aligned}$$

Adding (i) and (ii)

$$\|x+y\|^2 + \|x-y\|^2 = 2\|x\|^2 + 2\|y\|^2$$

That we have to prove.



Question18

For any two elements x, y in an inner product space V ,
 $|\langle x, y \rangle| \leq \|x\| \cdot \|y\|$



Prove:

If $x=y=0$ then $0=0$

Let at least one of x and y is not equal to zero

Let $|\langle x + \lambda y, x + \lambda y \rangle| \geq 0$ by definition

$$\langle x, x + \lambda y \rangle + \langle \lambda y, x + \lambda y \rangle$$

$$\langle x, x + \lambda y \rangle + \lambda \langle y, x + \lambda y \rangle$$

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Question 19

SHOW THAT T is uniquely determinal if the image $y_k = Te_k$ of n basis vectors $e_1, \dots, e_n$ are prescribed.

Prove:

$$y = Tx \quad ; y \in Y \quad \{b_1, \dots, b_r\}$$

$$y = \eta_1 b_1 + \eta_2 b_2 + \dots + \eta_r b_r$$

$$Te_k \in Y, \quad Te_k = \tau_{k1} b_1 + \tau_{k2} b_2 + \dots + \tau_{kr} b_r$$

$$Te_k = \sum_{j=1}^r \tau_{kj} b_j$$

$$y = \sum_{j=1}^r \eta_j b_j = \sum_{k=1}^n \xi_k Te_k = \sum_{k=1}^n \xi_k \sum_{j=1}^r \tau_{kj} b_j$$

Combining these two summation

$$y = \sum_{j=1}^r \left(\sum_{k=1}^n \tau_{kj} \xi_k \right) b_j$$

$$\eta_j = \sum_{k=1}^n \tau_{kj} \xi_k$$

The image $y = Tx = \sum \eta_j b_j$ of $x = \sum \xi_k Te_k$ can be obtained from

$$\eta_j = \sum_{k=1}^n \tau_{kj} \xi_k$$



Question 20

Sum and Scalar multiplication of Dual.

Prove:

as follows. The SUM $f_1 + f_2$ of two functionals f_1 and f_2 is the functional s whose value at every $x \in X$ is

$$s(x) = (f_1 + f_2)(x) = f_1(x) + f_2(x);$$

the Product αf of a scalar α and a functional f is the functional p whose value at $x \in X$ is

$$p(x) = (\alpha f)(x) = \alpha f(x).$$



Question 21

Define Inner Product Space:

The pair $(V, \langle \cdot, \cdot \rangle)$ is called an inner product space.

Define Appolonius Identity:

$$\|z - x\|^2 + \|z - y\|^2 = \frac{1}{2}\|x - y\|^2 + 2\left\|z - \frac{1}{2}(x + y)\right\|^2, \quad x, y, z \in V$$

Define Equal Operators:

Let T_1 and T_2 be two operators defined on equal i.e. $T_1 = T_2$ if they have the same

domain i.e. $D(T_1) = D(T_2)$



Question 22

Define Orthonormal System.

Orthonormal systems are those functions which are real or complex whose inner product with itself results in 1 and with other functions result in 0.

Define Polarization Identity.

For any x, y in complex inner product space

$$\langle x, y \rangle = \frac{1}{4} \left\{ \|x + y\|^2 - \|x - y\|^2 + i\|x + iy\|^2 - i\|x - iy\|^2 \right\}$$

We have to prove this complex inner product space.



Question 23

Define Bounded Linear Function.

Let X and Y be normed spaces and $T : D(T) \rightarrow Y$ a linear operator, where $D(T) \subset X$. The operator T is said to be bounded if there is a real number c such that for all $x \in D(T)$.

$$\|Tx\| \leq c\|x\|$$

Define Canonical Mapping.

$C : X \rightarrow X^{**}$ this mapping is called canonical mapping of X into X^{**} defined as $x \mapsto g_x$.



Question 24

Define Unitary Operator.

A bounded Linear Operator $T: H \rightarrow H$ on a Hilbert Space H is said to be Unitary Operator if T is bijective and $T^* = T^{-1}$.

Define Normal Operator.

A bounded Linear Operator $T: H \rightarrow H$ on a Hilbert Space H is said to be Unitary Operator if T is bijective and $TT^* = T^*T$

Define Extension of T .

Extension of T on a set $M \supset D(T)$ is defined as $\tilde{T}: M \rightarrow Y$ or
$$\tilde{T}x = Tx \text{ for all } x \in D(T)$$



Question 25

Write examples of Bounded Linear Operators

Identity Operator, Zero Operator, Differentiation Operator, Integral Operator

Define Self Adjoint Operator or Hermitian Operator.

A bounded Linear Operator $T: H \rightarrow H$ on a Hilbert Space H is said to be Self Adjoint Operator or Hermitian Operator if $T^* = T$.

Give an example of function which is not linear.

$\| \cdot \|: X \rightarrow R$ is functional but not linear.

Find basis of $R(T)$.

$$(\xi_1, \xi_2, -\xi_1, -\xi_2) = \xi_1(1, 0, -1) + \xi_2(0, 1, -1)$$

So the basis of $R(T) = \{(1, 0, -1), (0, 1, -1)\}$



Question 26

Give any two examples of Linear Functional.

Space $C[a, b]$ and Space l^2

Define Linear Functional:

A linear functional f is a linear operator with domain in a vector space X and range in the scalar field K of X ; thus $f: D(f) \rightarrow K$. where $K=\mathbb{R}$ if X is real and $K=\mathbb{C}$ if X is complex.

Define Bounded Linear Extension:

$T: D(T) \rightarrow Y$ be a bounded linear operator, where $D(T)$ lies in a normed space X and Y is a Banach Space then T has an extension $\tilde{T}: \overline{D(T)} \rightarrow Y$. Where $\tilde{T}$ is a bounded linear operator of norm.

$$\| \tilde{T} \| = \| T \|$$



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